

# **5.MICRODEBRIDER**

## **TECHNICAL SPECIFICATION**

### **INTEGRATED POWER CONSOLE WITH ENT MICRODEBRIDER SYSTEM**

#### **Intended Application:**

ENT, Rhinology, Functional Endoscopic Sinus Surgery (FESS), Adenoidectomy, Turbinate Surgery, Polypectomy, Laryngeal Procedures, Otology, Neurotology, Skull Base and other applicable ENT procedures.

#### **A. Main Console**

| <b>Sl. No.</b> | <b>Parameter</b>                  | <b>Minimum Tender Requirement</b>                                                                          |
|----------------|-----------------------------------|------------------------------------------------------------------------------------------------------------|
| 1              | Type                              | Advanced microprocessor-controlled integrated ENT surgical power console                                   |
| 2              | Application                       | Cutting, removal, drilling and sawing of soft tissue, hard tissue and bone                                 |
| 3              | Surgical Specialty                | ENT, Otology, Neurotology, Rhinology, Sinus, Skull Base and Head & Neck surgery                            |
| 4              | Control                           | Touchscreen-based user interface                                                                           |
| 5              | Handpiece Compatibility           | Shall support multiple compatible ENT powered handpieces                                                   |
| 6              | Simultaneous Handpiece Connection | Minimum <b>4 handpieces</b> connectable to the console                                                     |
| 7              | Microdebrider                     | Compatible with high-speed ENT microdebrider handpieces                                                    |
| 8              | High-Speed Microdebrider          | Should support microdebrider speeds up to <b>30,000 RPM or better</b> with compatible high-speed handpiece |
| 9              | Drill                             | Compatible with high-speed otologic/ENT drill handpiece                                                    |
| 10             | Burs                              | Compatible with application-specific high-speed burs                                                       |
| 11             | Cutting Blades                    | Compatible with a comprehensive range of straight, curved and specialty blades                             |
| 12             | Speed Control                     | Adjustable speed control according to handpiece and surgical application                                   |
| 13             | Rotation Control                  | Forward and reverse rotation where applicable                                                              |
| 14             | Oscillation                       | Oscillation mode should be available for suitable microdebrider applications                               |
| 15             | Irrigation                        | Integrated suction-irrigation system                                                                       |
| 16             | Irrigation Flow                   | Adjustable irrigation flow, preferably <b>0.5–100 mL/min or better</b>                                     |

| Sl. No. | Parameter                 | Minimum Tender Requirement                                                                                |
|---------|---------------------------|-----------------------------------------------------------------------------------------------------------|
| 17      | Remote Irrigation Control | Sterile-field remote control for irrigation ON/OFF, tip size and flow-rate adjustment                     |
| 18      | Foot Control              | Multifunction foot pedal for speed, activation and other programmable controls                            |
| 19      | Foot Pedal                | <b>Minimum 3-button multifunction foot pedal</b>                                                          |
| 20      | Suction                   | Integrated suction capability through compatible microdebrider handpieces                                 |
| 21      | Display                   | Colour touchscreen display with clear visualization of selected handpiece, speed and operating parameters |
| 22      | Presets                   | User-configurable/customizable default settings                                                           |
| 23      | Safety                    | System shall provide appropriate safety controls to prevent inadvertent handpiece activation              |
| 24      | Handpiece Recognition     | Automatic or guided identification/configuration of compatible handpieces, where applicable               |
| 25      | Power Supply              | <b>220–240 V AC, 50/60 Hz</b> or suitable hospital mains supply                                           |
| 26      | System Integration        | Compatible with other manufacturer-supported ENT surgical equipment and accessories                       |
| 27      | Navigation Compatibility  | Should support compatible image-guided surgery/navigation instruments where applicable                    |
| 28      | Lens Cleaning             | Should be compatible with powered endoscopic lens-cleaning accessories, where required                    |

## B. MICRODEBRIDER HANDPIECE

| Parameter           | Minimum Tender Requirement                                                                    |
|---------------------|-----------------------------------------------------------------------------------------------|
| Type                | High-performance ENT microdebrider handpiece                                                  |
| Application         | FESS, sinus surgery, polypectomy, turbinate reduction, adenoidectomy and other ENT procedures |
| Motor               | Direct-drive or equivalent high-performance motor                                             |
| Maximum Speed       | <b>Minimum 30,000 RPM</b> with compatible high-speed blades/burs                              |
| Suction             | Integrated straight-through suction                                                           |
| Rotation            | Variable rotation control                                                                     |
| Tip Control         | Rotatable cutting tip preferred                                                               |
| Tip Rotation        | <b>360° independently rotatable tip</b> preferred                                             |
| Ergonomics          | Lightweight and balanced design suitable for prolonged surgery                                |
| Blade Compatibility | Wide range of disposable blades and burs                                                      |
| Blade Types         | Straight, curved, angled and specialty blades                                                 |

| Parameter          | Minimum Tender Requirement                                                  |
|--------------------|-----------------------------------------------------------------------------|
| Burr Compatibility | High-speed burs for frontal sinus and skull-base procedures                 |
| Navigation         | Navigation-compatible blades/burs should be available                       |
| Sterilization      | Reusable handpiece shall be compatible with validated sterilization process |
| Connection         | Quick and secure connection to console                                      |
| Control            | Handpiece and foot-pedal control                                            |

## C. CURVED / ROTATING MICRODEBRIDER

The system should preferably include a curved microdebrider handpiece with:

- Ergonomic design
- Integrated suction
- Rotatable distal cutting tip
- Independent tip rotation without rotating the shaft
- 360° cutting capability
- Finger-wheel tip control
- Compatibility with curved blades
- Suitable for frontal sinus and maxillary sinus procedures
- Suitable for polyp and mucin removal
- The tip of a curved blade to rotate **360° without shaft rotation**, providing enhanced access in frontal and maxillary sinus procedures.

## D. IRRIGATION SYSTEM

The system shall include:

1. Integrated irrigation pump.
2. Adjustable irrigation flow.
3. Sterile-field remote irrigation control.
4. Irrigation ON/OFF control.
5. Adjustable flow according to blade/tip size.
6. Compatible sterile irrigation tubing.
7. Automatic or preset irrigation settings for different tips.
8. Continuous irrigation capability.
9. Pulse-flow irrigation capability, where supported.
10. Irrigation status display on console.
11. Specifies continuous irrigation control from approximately **0.5–100 cc/min** and pulse-flow irrigation at lower flow settings, depending on configuration.

## E. FOOT CONTROL

The system shall be supplied with:

- Multifunction foot pedal – **1 No.**
- Minimum 3 programmable buttons
- Non-slip design
- Hands-free control of handpiece activation
- Speed control
- Forward/reverse or oscillation control where applicable
- Irrigation control
- Safe locking mechanism
- Long, flexible connecting cable
- Multifunction foot pedal provides three buttons for ENT surgical control.

## F. REQUIRED ACCESSORIES

The complete system shall include:

1. Integrated Power Console – **1 No.**
2. Multifunction foot pedal – **1 No.**
3. IPC equipment cart/trolley – **1 No.**
4. High-speed microdebrider handpiece – **1 No.**
5. Curved/rotatable microdebrider handpiece – **1 No.**
6. High-speed ENT drill handpiece – **1 No.**, if required
7. Sterilization tray – **1 No.**
8. Irrigation pump system – integrated
9. Remote irrigation control – **1 No.**
10. Suction-irrigation tubing sets – initial supply
11. Adult straight microdebrider blades – assorted
12. Curved microdebrider blades – assorted
13. Pediatric microdebrider blades – assorted
14. High-speed burs – assorted
15. Frontal sinus burs – assorted
16. Skull-base compatible burs – assorted
17. Navigation-compatible blades/burs – if required
18. Power cord – **1 No.**
19. User manual – **1 set**
20. Service manual – **1 set**
21. Installation and commissioning
22. Comprehensive user training

The exact blade and burr assortment should be finalized based on the ENT surgeon's clinical requirement because Companies offers a broad portfolio of application-specific blades and burs.

## G. SAFETY REQUIREMENTS

The system shall have:

- Prevention of inadvertent activation.
- Audible and visual activation indication.
- Foot-pedal safety control.
- Overload protection.
- Motor protection.
- Appropriate electrical safety protection.
- Handpiece connection verification.
- Error/fault indication.
- Safe shutdown in case of system fault.
- Protection against accidental blade activation.
- Compliance with applicable medical electrical safety standards.

## H. WARRANTY & SERVICE

| Parameter              | Tender Requirement                                                   |
|------------------------|----------------------------------------------------------------------|
| Warranty               | Minimum 3 <b>years comprehensive warranty</b>                        |
| Service                | Authorized service support in India                                  |
| Response Time          | Preferably within <b>24 hours</b>                                    |
| Preventive Maintenance | Minimum 4 visits per year during warranty                            |
| Spare Parts            | Availability for minimum 7–10 years                                  |
| Handpiece Service      | Service/repair support for reusable handpieces                       |
| Training               | Training for ENT surgeons, OT nurses and Biomedical Engineering team |
| Installation           | Installation, testing and commissioning included                     |
| Demonstration          | Live demonstration if required by technical committee                |
| Documentation          | User manual, service manual, IFU and sterilization guidelines        |